

Entity- Relationship Model (ER Model)

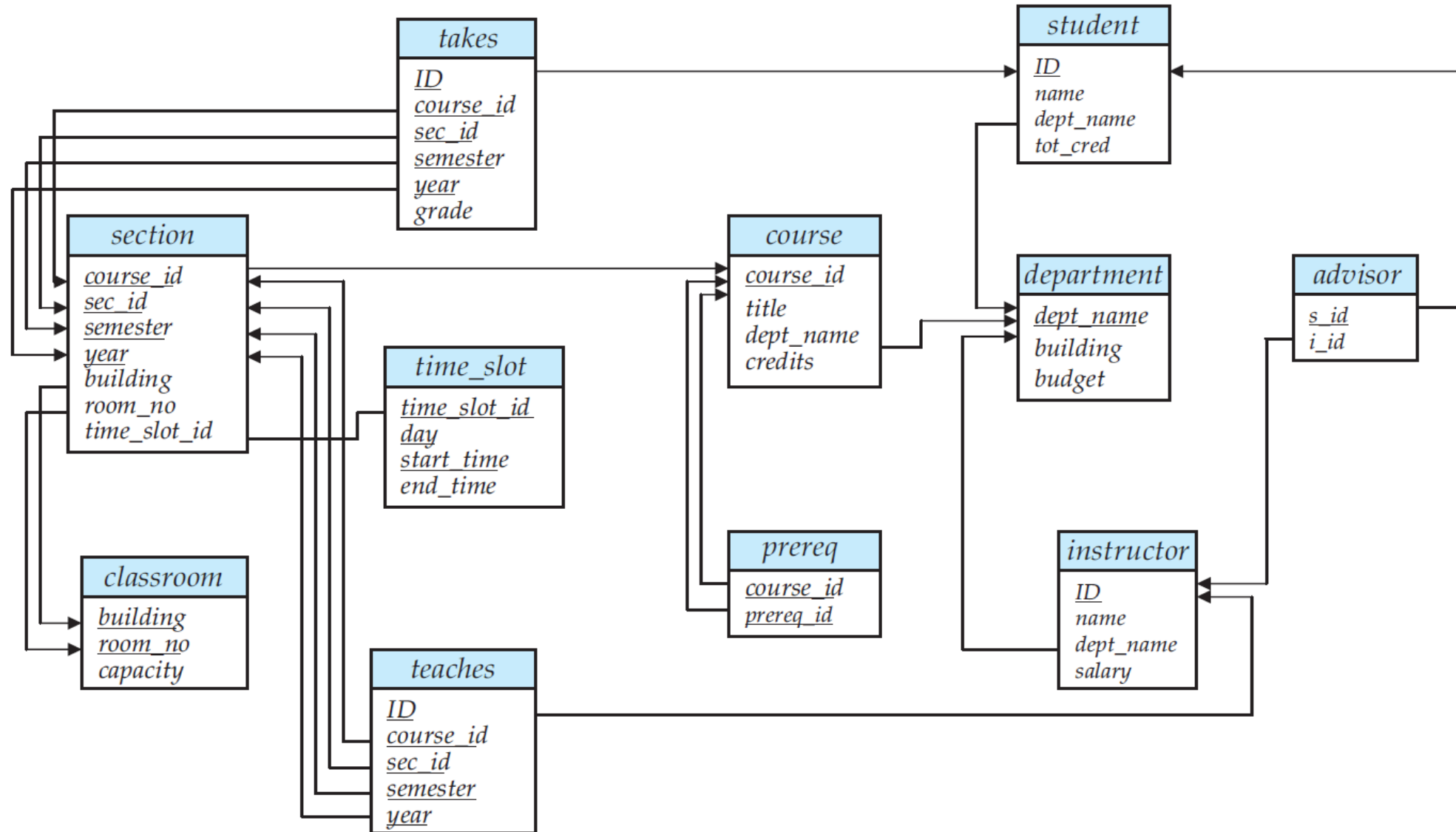


Agenda

- Design Process
- Modeling
- Constraints
- E-R Diagram
- Design Issues
- Weak Entity Sets
- Extended E-R Features
- Reduction to Relation Schemas
- Database Design
- UML

Schema Diagram

University Database



Database Design

DB Design Process:

- **Phase I:** user requirements
- **Phase II:** data model selection and requirement translation into a conceptual schema

Database Design Phases

- Designs:

1. Logical Design

- Deciding on the database schema,
- e.g., find a “good” collection of relation schemas.
 - **Business decision;**
 - **Computer Science decision;**

2. Physical Design

- Deciding on the physical layout of the database.

Design Approaches

- **Entity-Relationship Model**

- Models an enterprise as a collection of **entities** and **relationships**.
 - **Entity**: a “*thing*” or “*object*” in the enterprise that is distinguishable from other objects
 - **Relationship**: an association among several entities

ER model - Database Modeling

- **ER data model**
 - Developed to facilitate database design
 - The basic concepts:
 - **Entity sets,**
 - **Relationship sets,**
 - **Attributes.**

Entity Sets

- **Entity**: an object that exists and is distinguishable from other objects.
 - e.g. specific person, company, event, plant, etc.
- **Entity set**: a set of entities of the same type that share the same properties.
 - e.g. set of all persons, companies, trees, holidays, etc.

- Example:

instructor = (ID, name, street, city, salary)

course = (course_id, title, credits)

Entity Sets

76766	Crick
45565	Katz
10101	Srinivasan
98345	Kim
76543	Singh
22222	Einstein

instructor

98988	Tanaka
12345	Shankar
00128	Zhang
76543	Brown
76653	Aoi
23121	Chavez
44553	Peltier

student

Relationship Sets

- **Relationship:** an association among several entities

e.g.:

44553 (Peltier)
student entity

advisor
relationship set

22222 (Einstein)
instructor entity

- **Relationship set:** mathematical relation among $n \geq 2$ entities, each taken from entity sets:

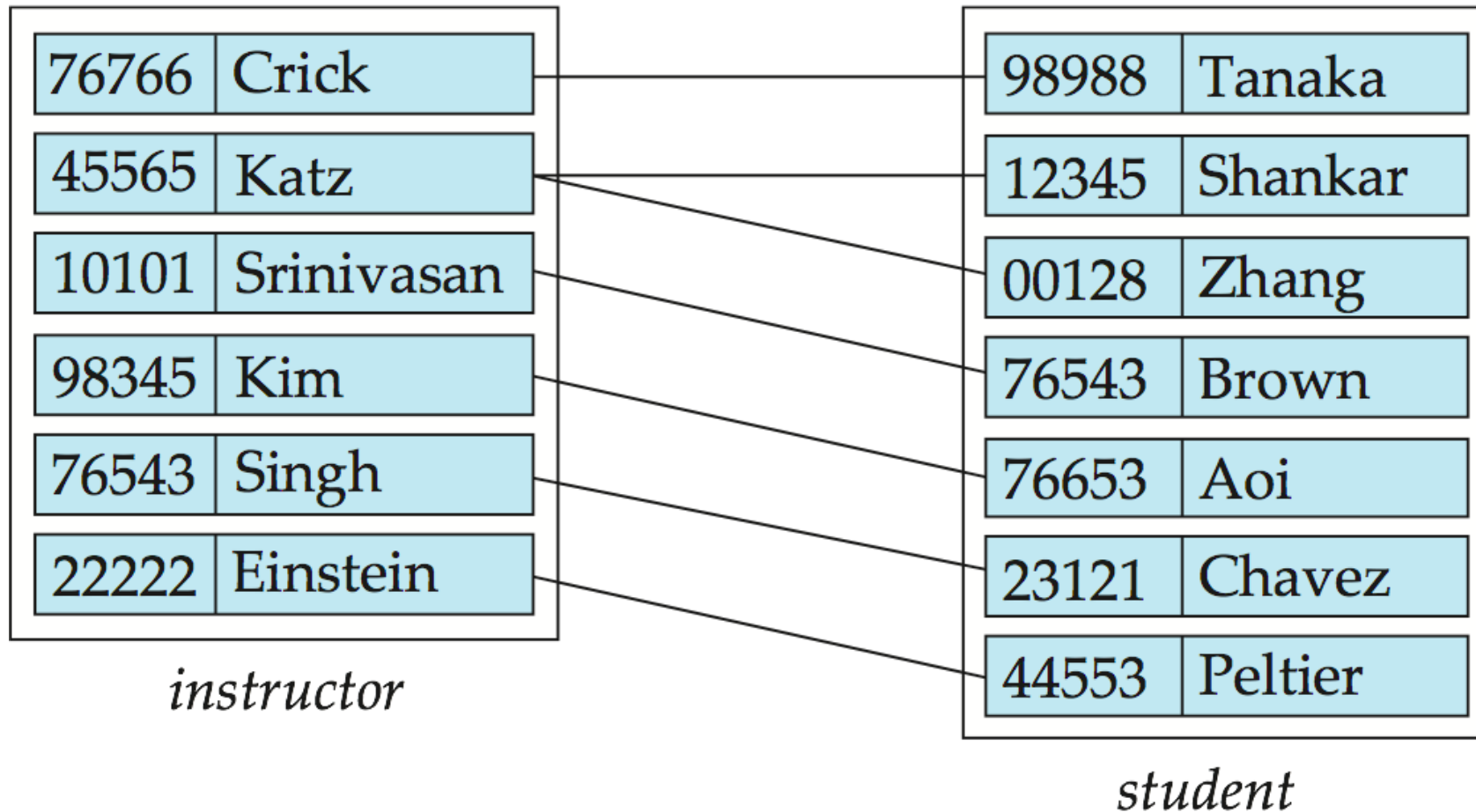
$$\{(e_1, e_2, \dots, e_n) | e_1 \in E_1, e_2 \in E_2, \dots, e_n \in E_n\}$$

- $(e_1, e_2, \dots, e_n)$ is a relationship
 - Example:

$$(44553, 22222) \in \textit{advisor}$$

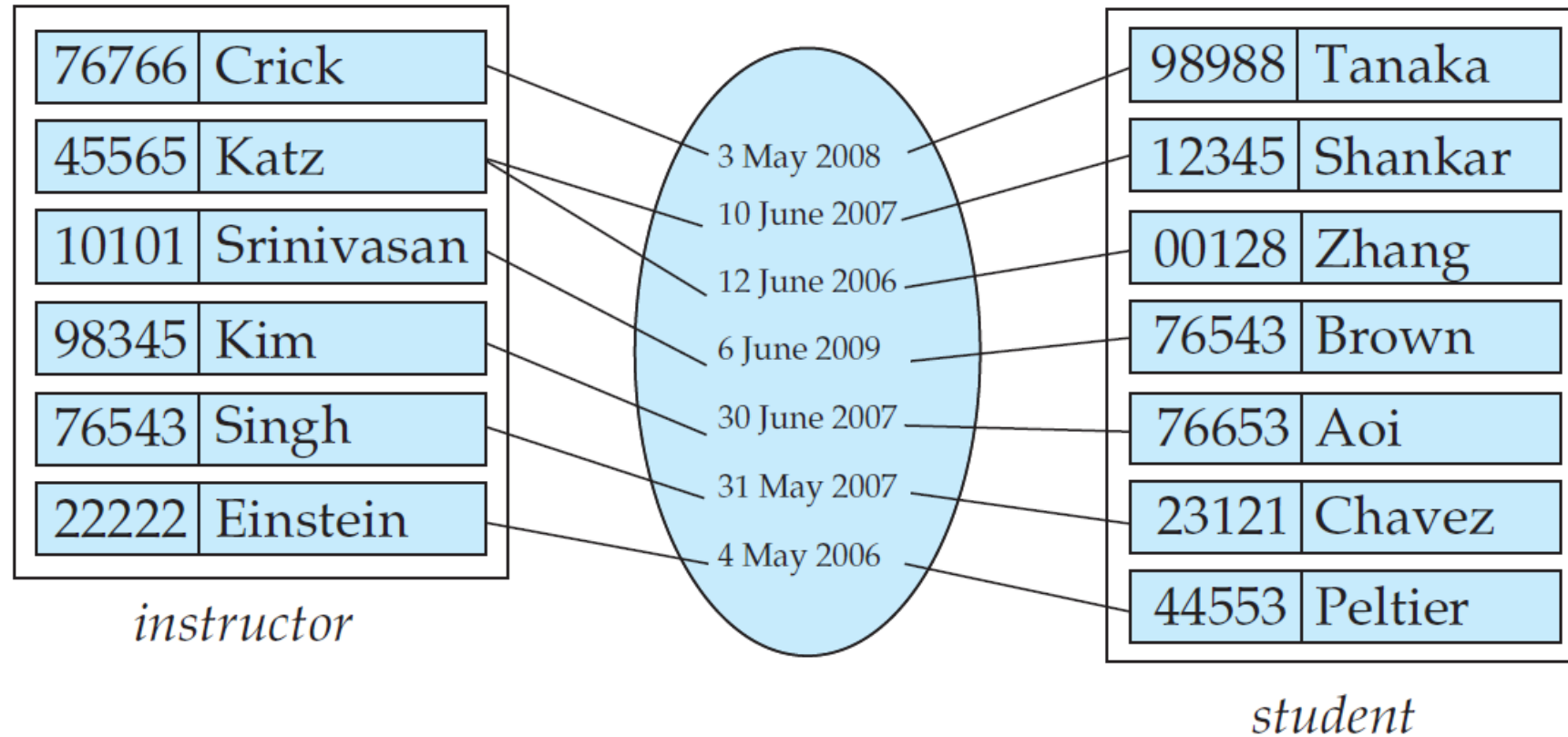
Relationship Sets

- Relationship Set *advisor*



Relationship Sets

- An attribute associated with a relationship set:



Degree of a Relationship Set

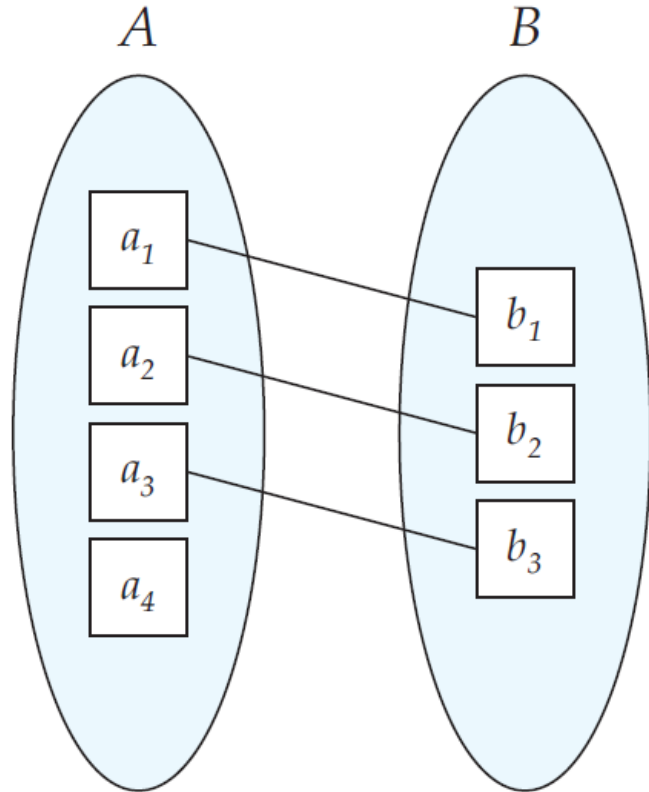
- **Binary relationship**
 - Involve two entity sets (or degree two).
- e.g.: *students* work on research *projects* under the guidance of an *instructor*.

Relationship Set

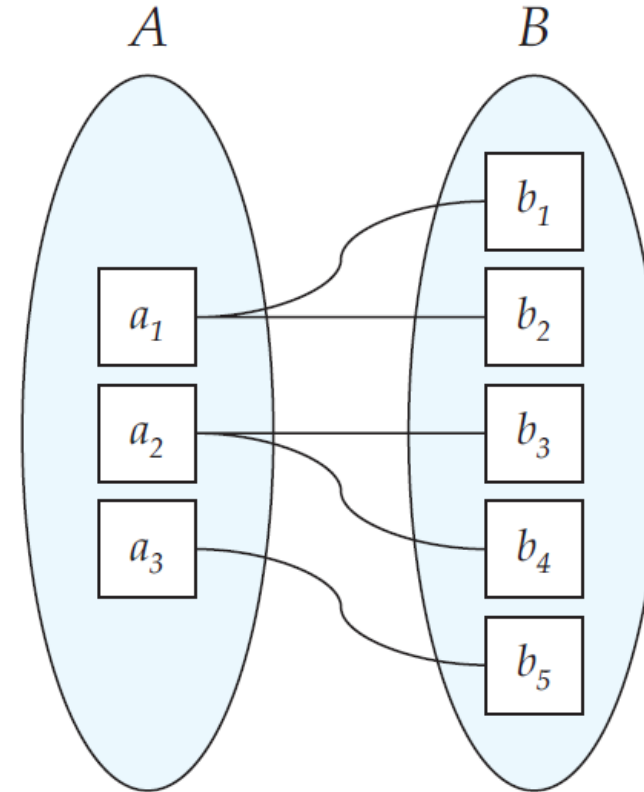
Mapping Cardinality Constraints

- Express the number of entities to which another entity can be associated via a relationship set.
- Types:
 - **One-to-one**
 - **One-to-many**
 - **Many-to-one**
 - **Many-to-many**

Mapping Cardinalities

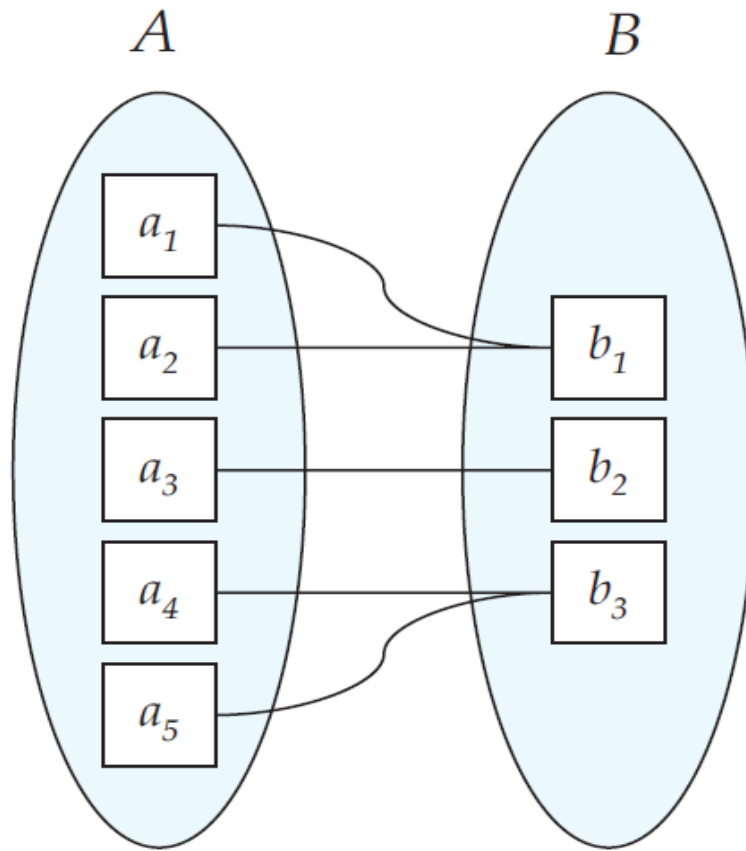


One-to-one

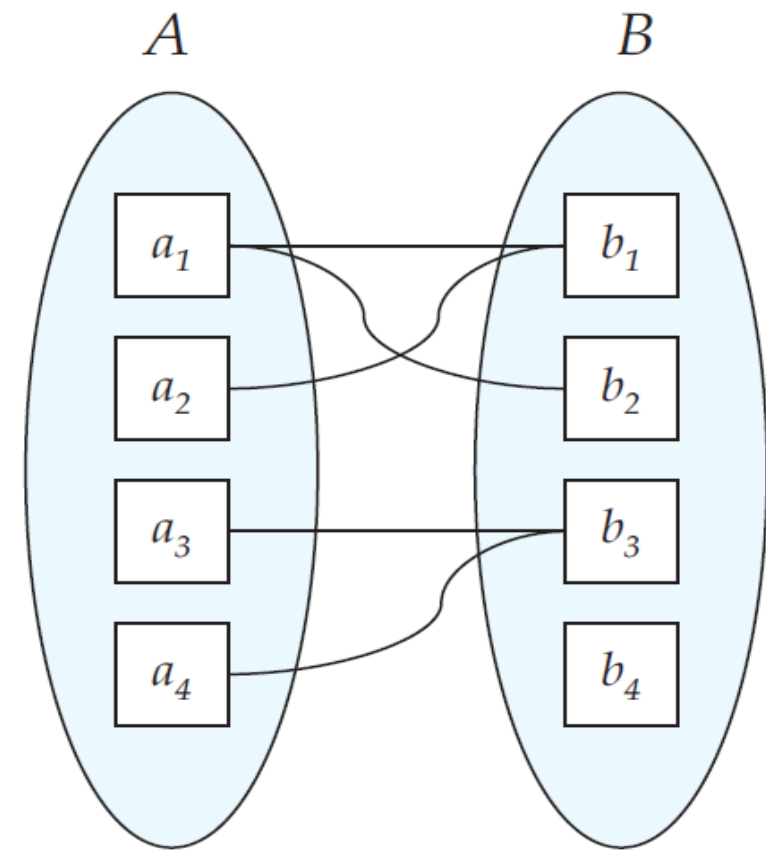


One-to-many

Mapping Cardinalities



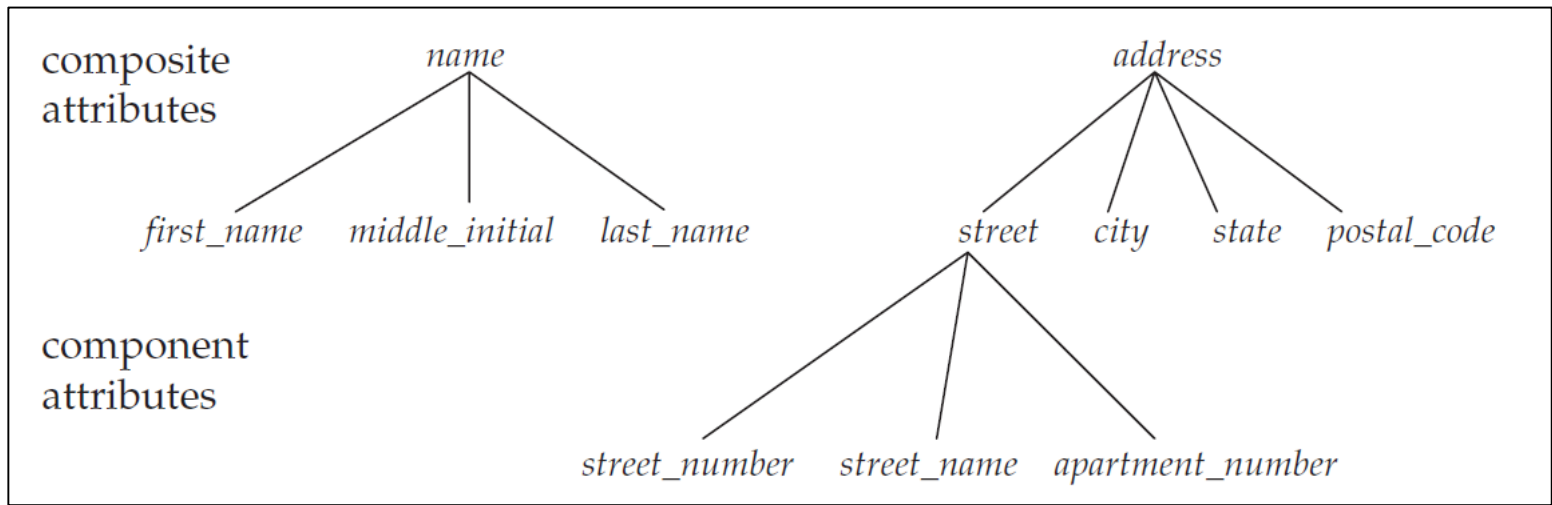
Many-to-one



Many-to-many

Attributes

- **Types:**
 - **Simple**
 - **Composite**
 - **Single-valued**
 - **Multivalued**
 - **Derived**



Redundant Attributes

- *Instructor:*

ID, name, dept_name, salary

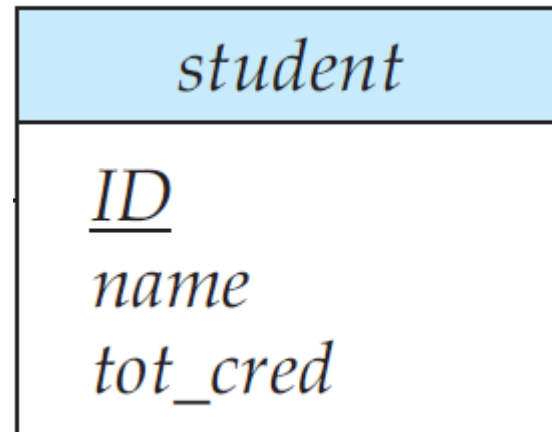
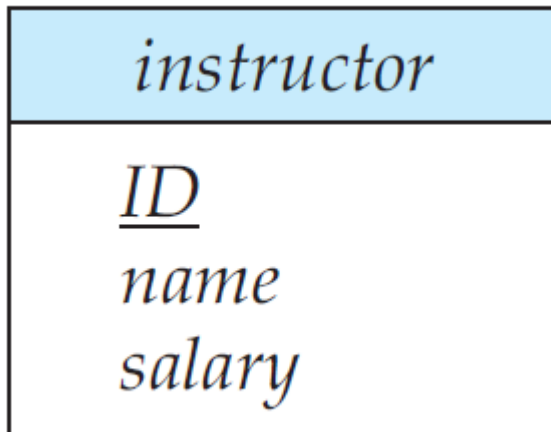
- *department:*

dept_name, building, budget

- **Redundant attribute:** *dept_name*

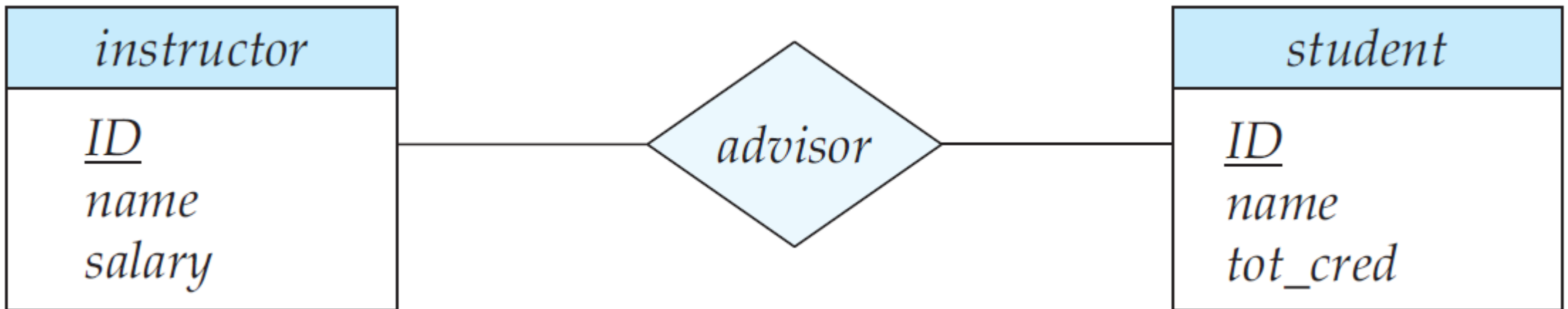
Entity Sets

- Graphic representation:
 - **Entity sets:** Rectangles
 - **Attributes:** listed inside entity rectangle
 - **Primary key:** Underline indicates

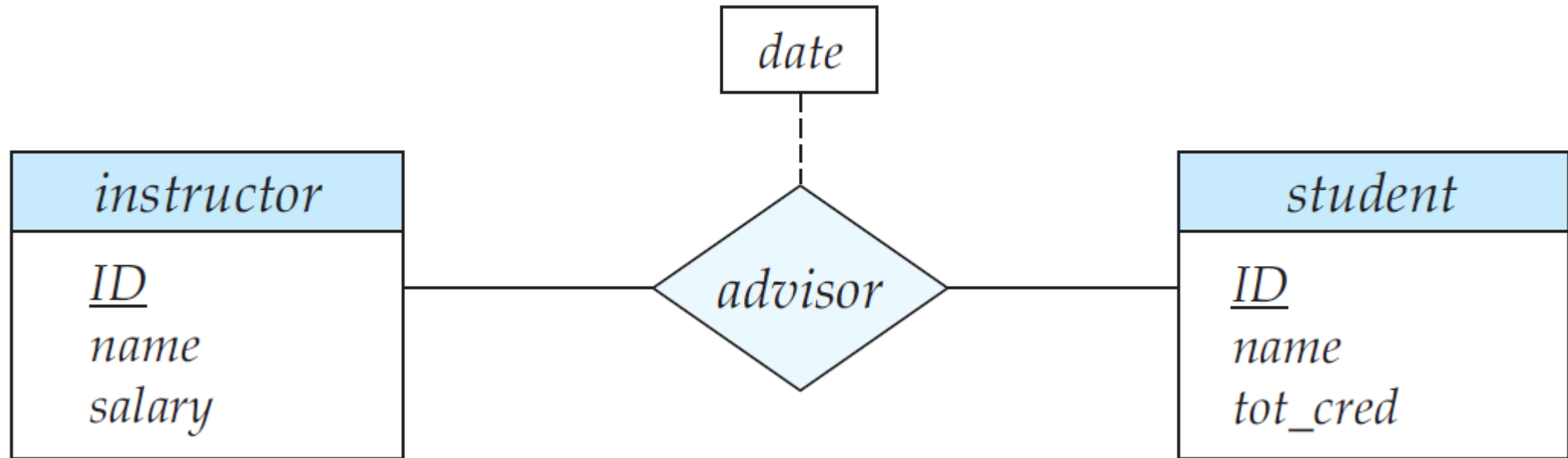


Relationship Sets

- **Relationship sets:** Diamonds.

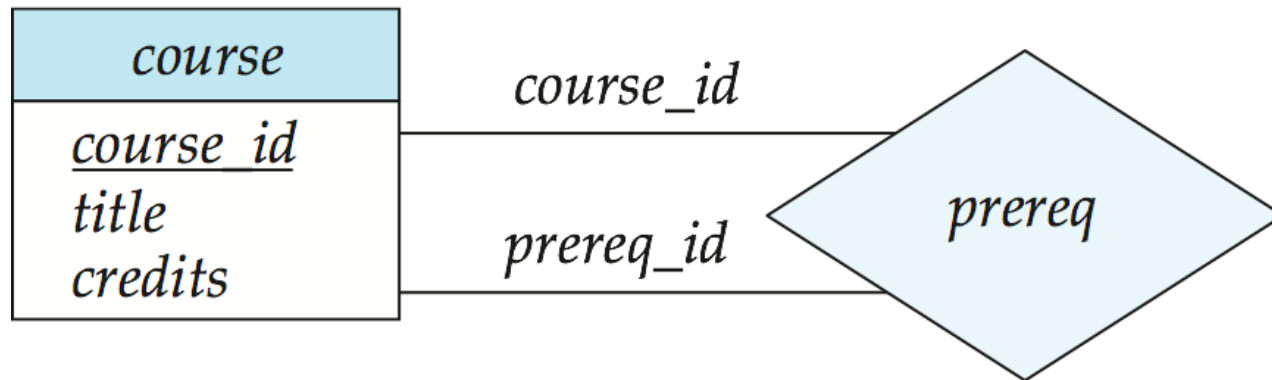


Relationship Sets with Attributes



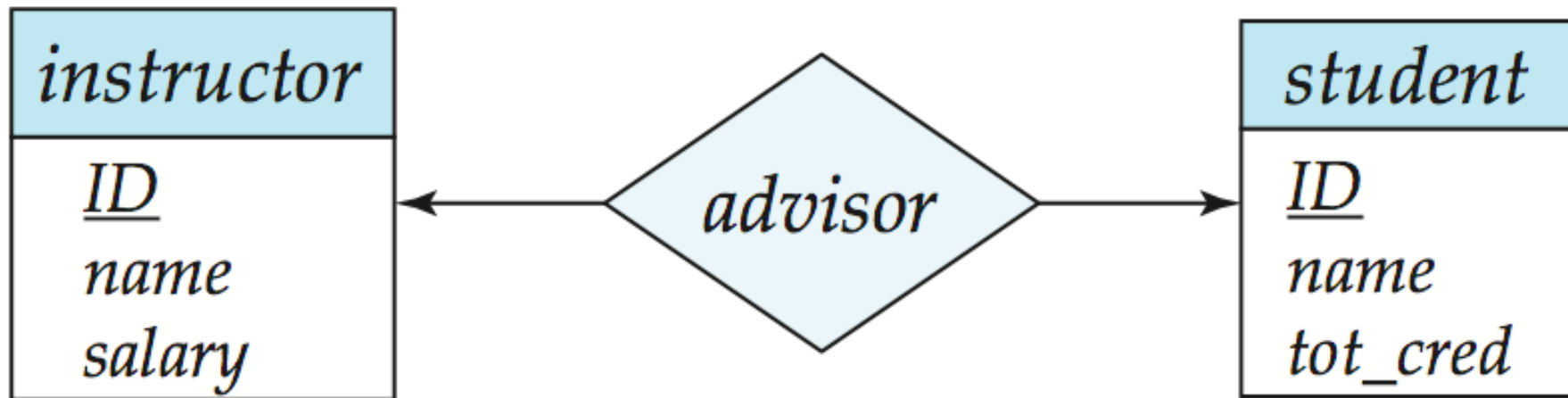
Roles

- Each occurrence of an entity set plays a “**role**” in the relationship
 - e.g., “*course_id*” and “*prereq_id*”.



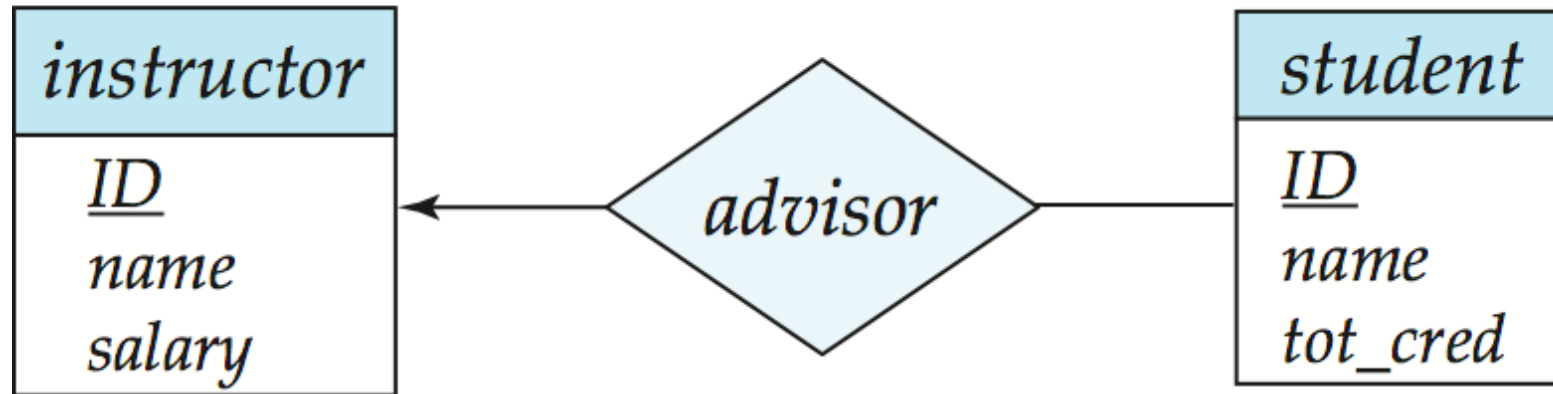
Cardinality Constraints

- **Cardinality constraints:**
 - **One:** →
 - **Many:** —
- One-to-one relationship between an *instructor* and a *student* :



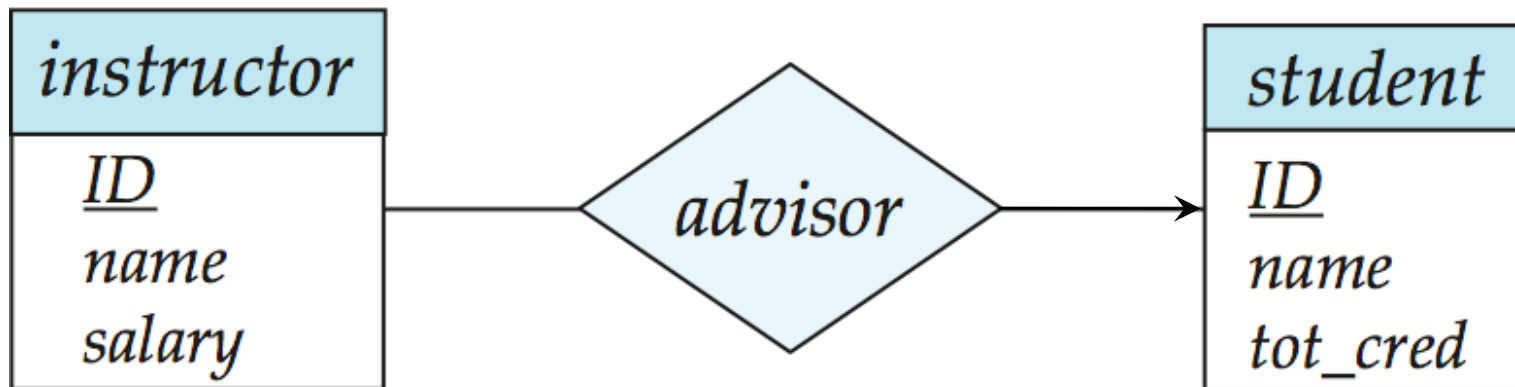
One-to-Many Relationship

- One-to-many relationship between an *instructor* and a *student*

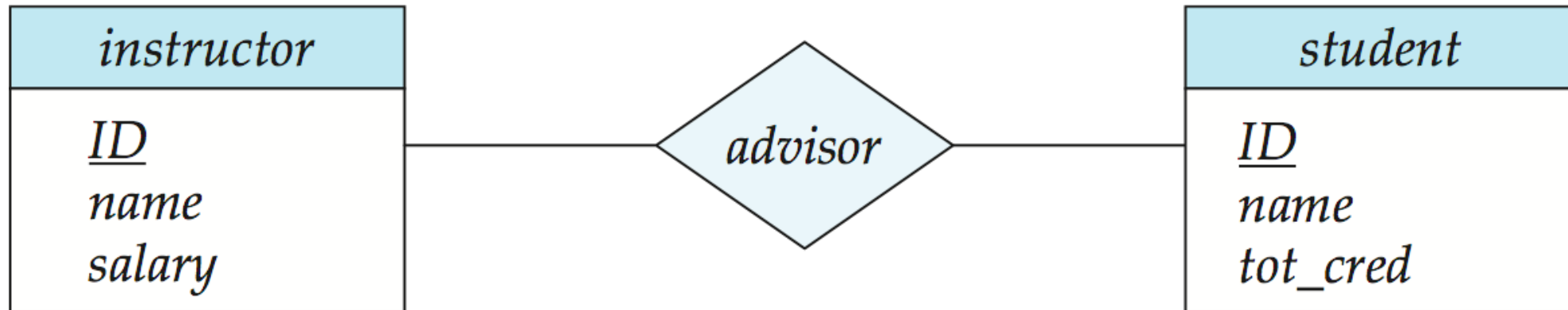


Many-to-One Relationships

- Many-to-one relationship between an *instructor* and a *student*,

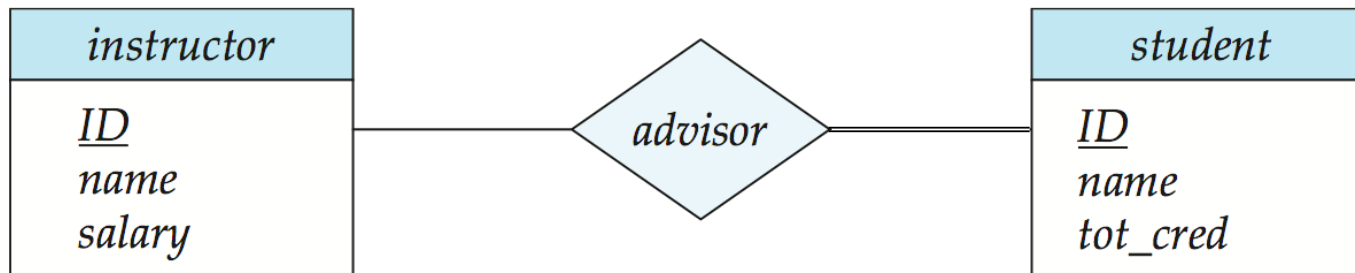


Many-to-Many Relationship



Total and Partial Participation

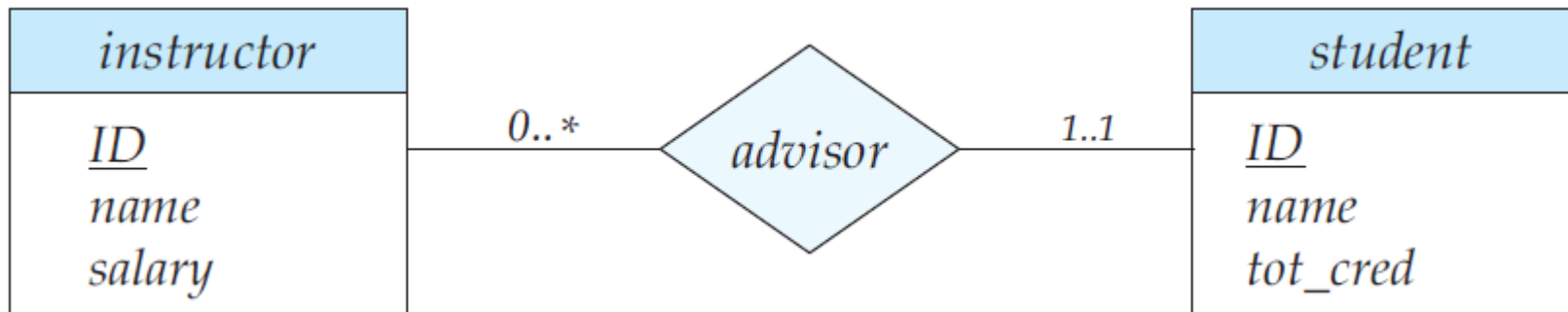
- **Total participation:** =
 - Every entity in the entity set participates in at least one relationship in the relationship set



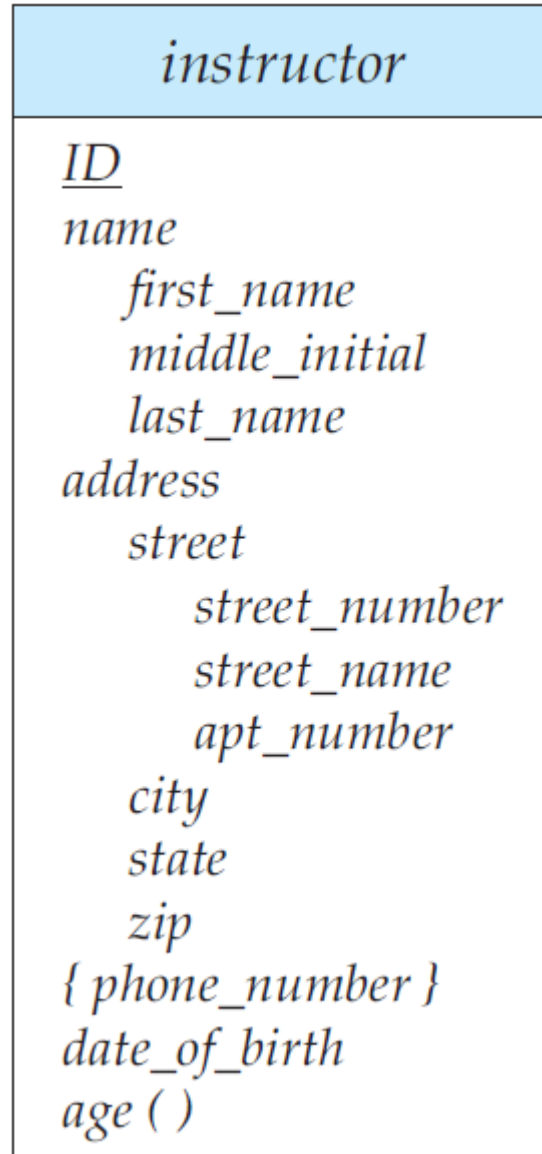
- **Partial participation:** some entities may not participate in any relationship in the relationship set

Notation for Expressing More Complex Constraints

- A line with an associated minimum and maximum cardinality,
- Notation: $l..h$
 - l : the minimum
 - $l = 1$: total participation.
 - h : the maximum cardinality
 - $h = 1$: the entity participates in at most one relationship
 - $h = *$: no limit.

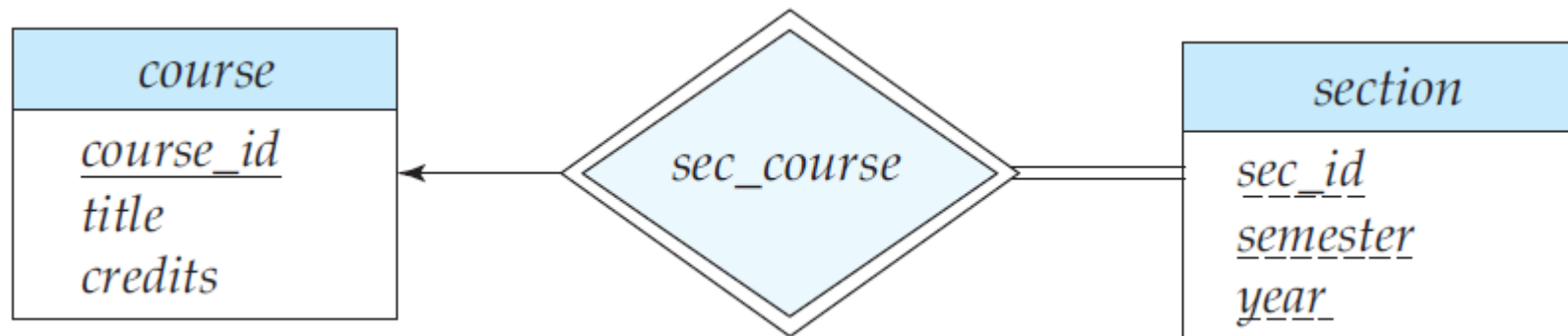


Notation to Express Entity with Complex Attributes



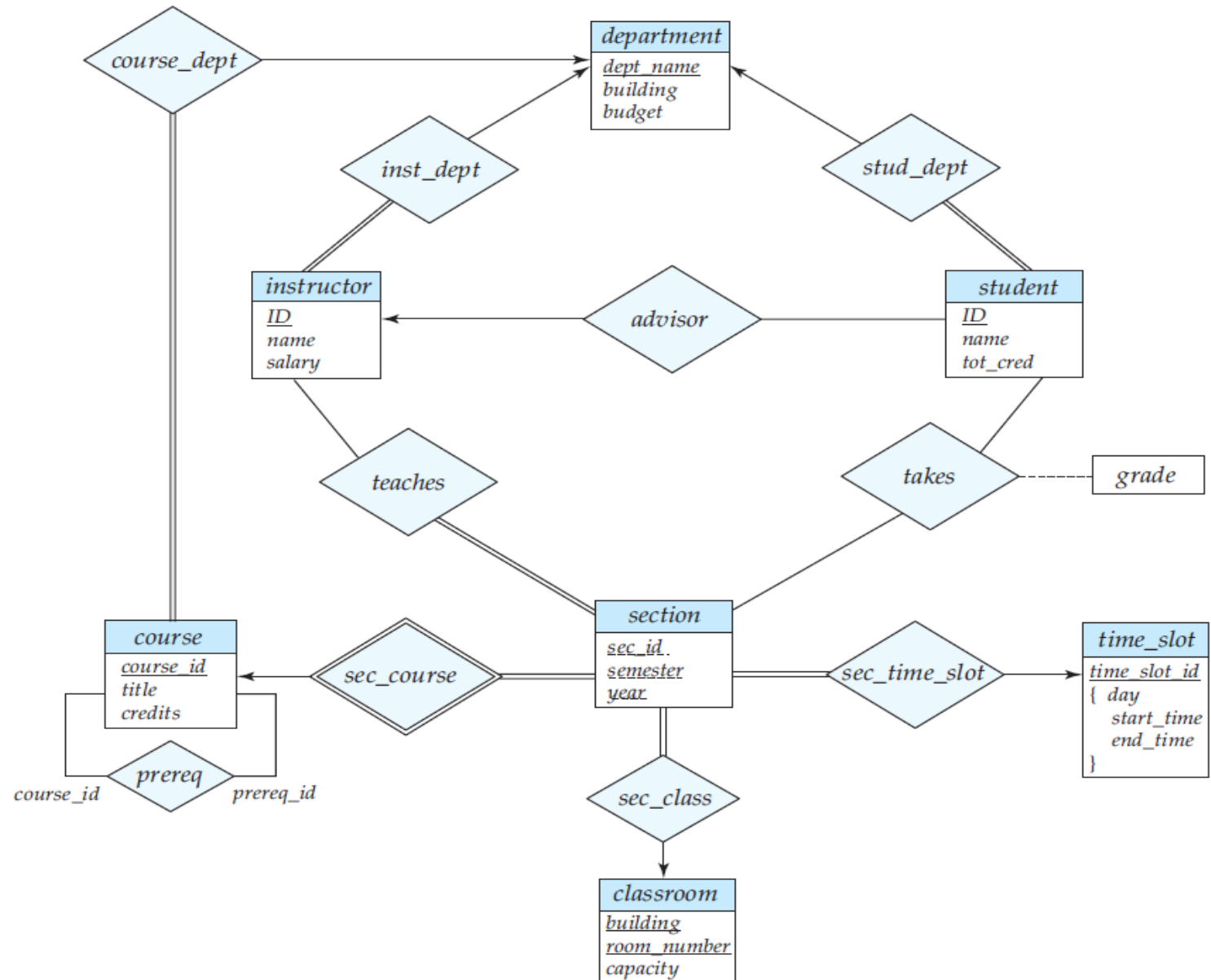
Expressing Weak Entity Sets

- **Weak entity:**
 - an entity that cannot be uniquely identified by its attributes alone;
 - Weak entity: **double line**.
 - Weak entity set: **dashed line**.



E-R Diagram

University Enterprise



Representing Entity Sets

- A strong entity set reduces to a schema with the same attributes

student(ID, name, tot_cred)

- A weak entity set becomes a table that includes a column for the primary key of the identifying strong entity set

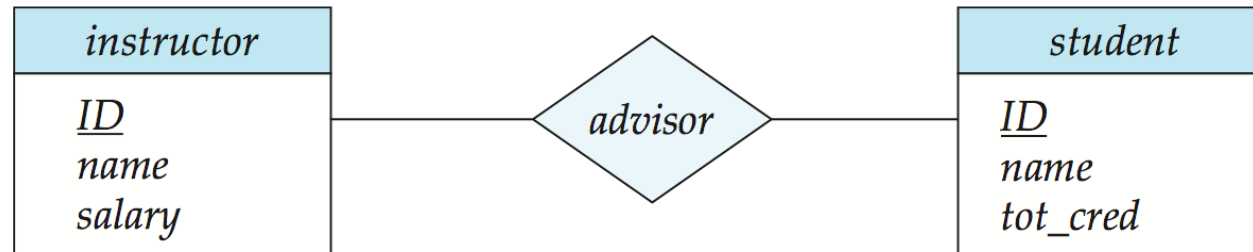
section (course_id, sec_id, sem, year)



Representing Relationship Sets

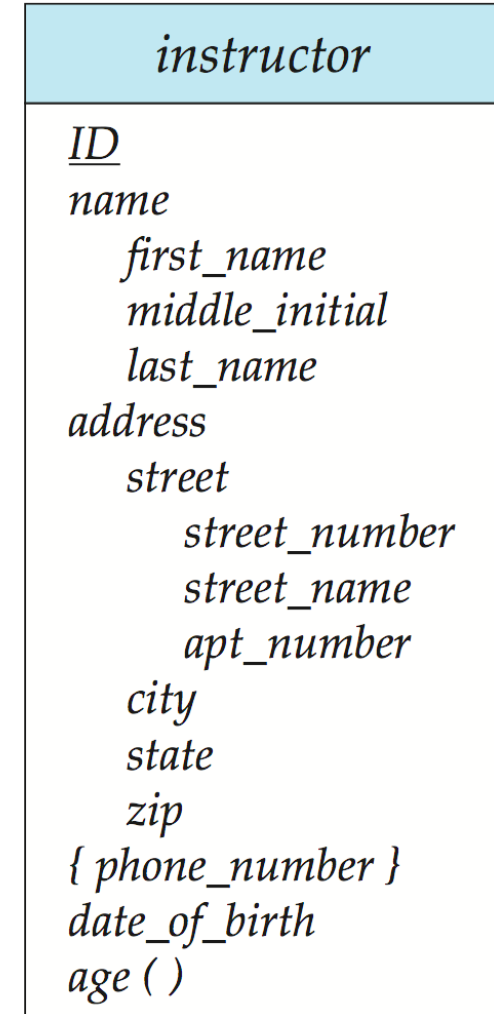
- Example: schema for relationship set *advisor*

$advisor = (s_id, i_id)$



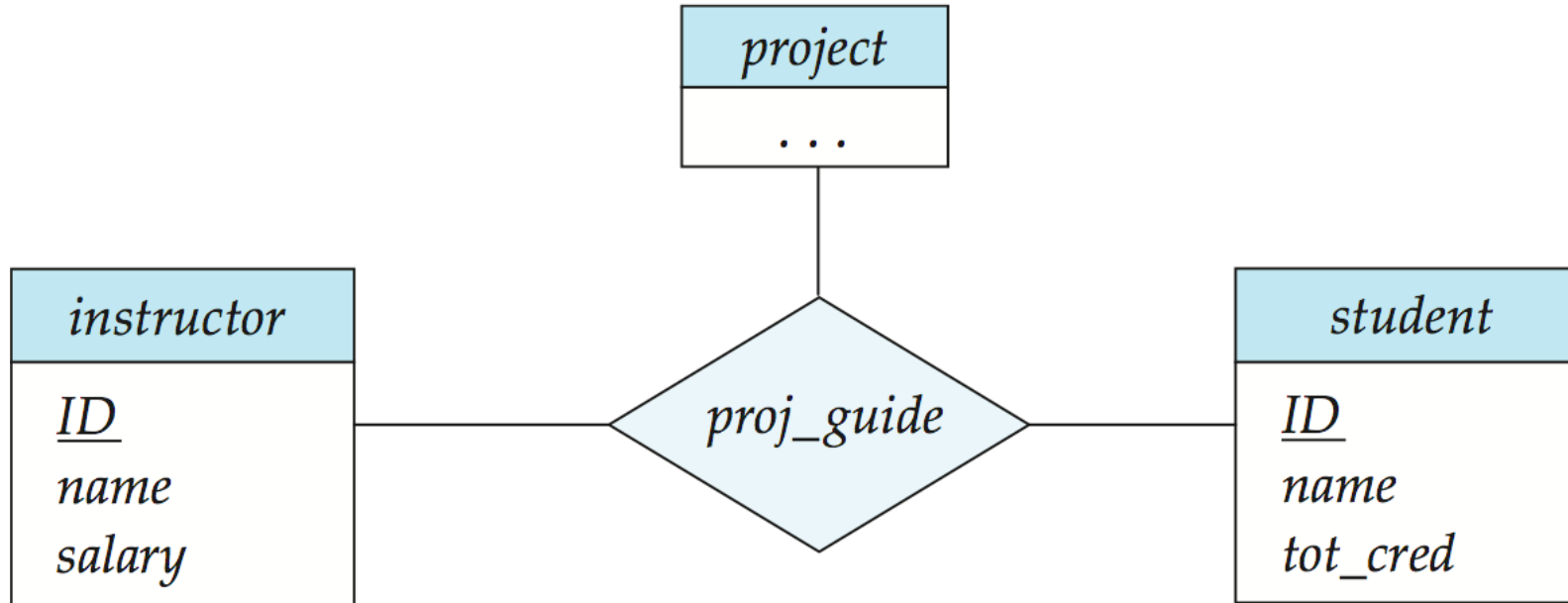
Representation of Entity Sets with Composite Attributes

- Composite attributes can be flattened.

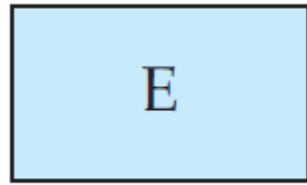


Non-binary Relationship Sets

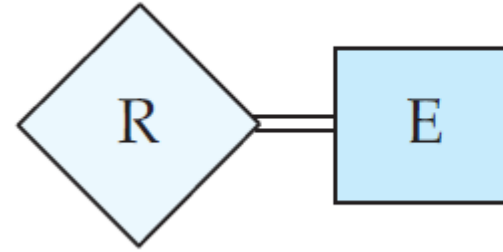
- E-R Diagram with a Ternary Relationship



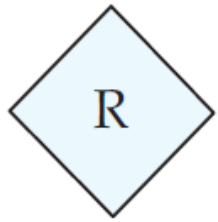
Symbols Used in E-R Notation



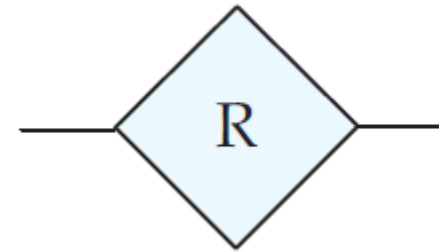
entity set



total participation
of entity set in
relationship



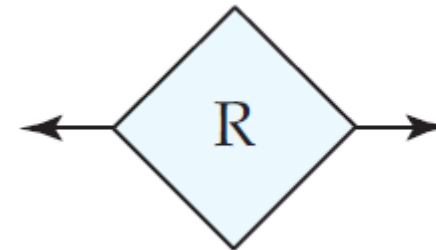
relationship set



many-to-many
relationship

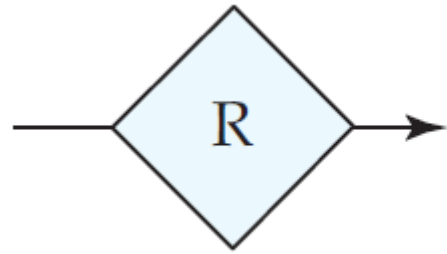


identifying
relationship set
for weak entity set

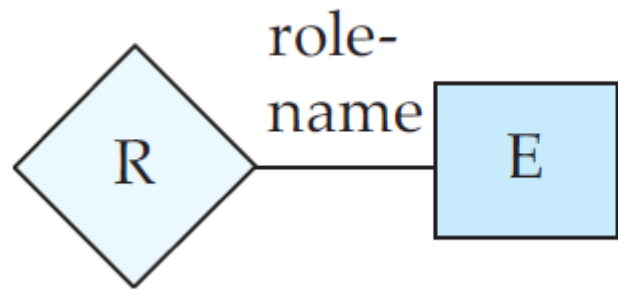


one-to-one
relationship

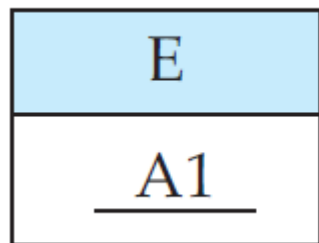
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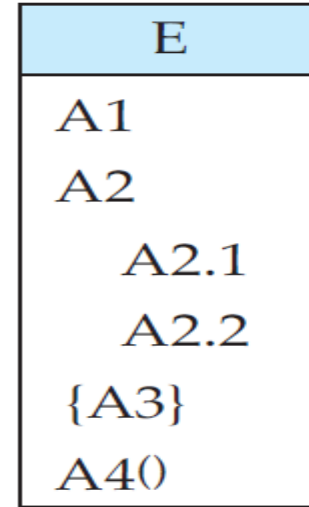
many-to-one
relationship



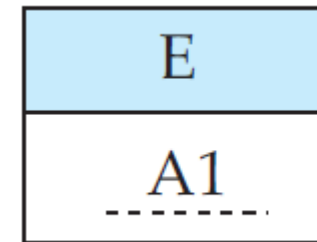
role indicator



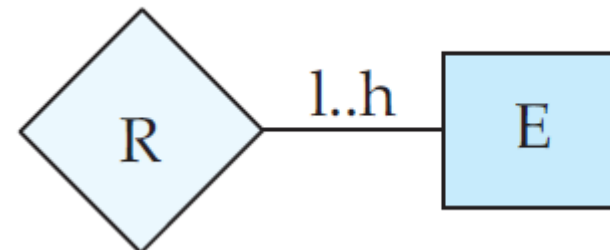
primary key



attributes:
simple (A1),
composite (A2) and
multivalued (A3)
derived (A4)



discriminating
attribute of
weak entity set



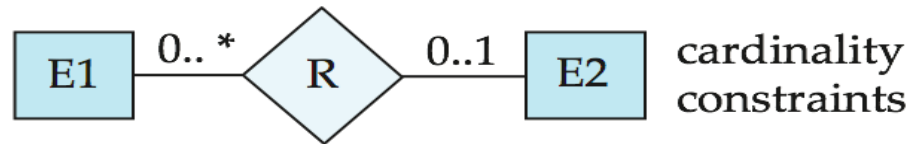
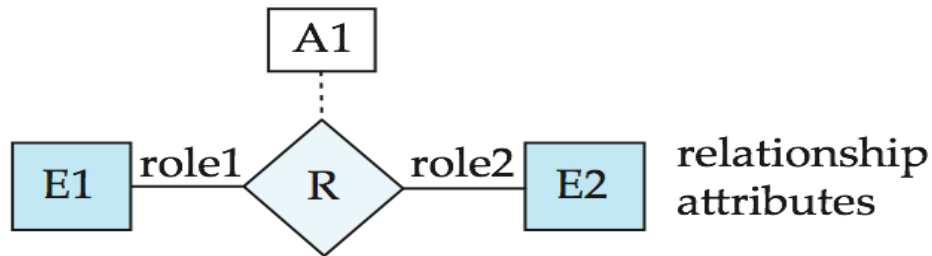
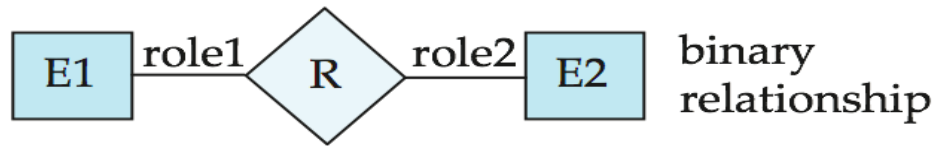
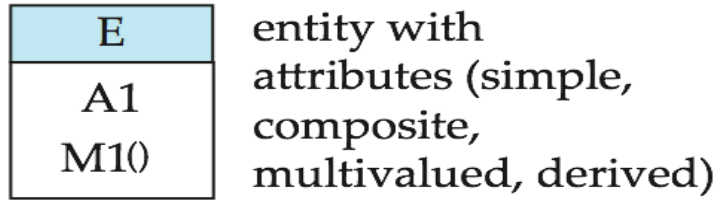
cardinality
limits

Unified Modeling Language (UML)

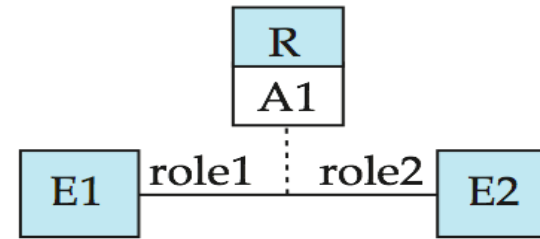
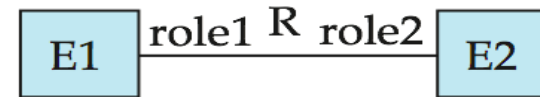
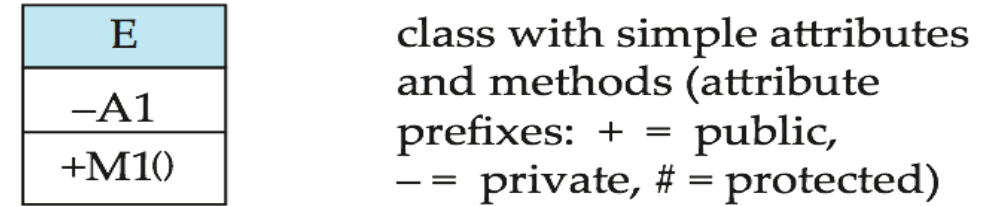
- **UML:**
 - A standard pictorial language for specifying, visualizing, constructing, and documenting the artifacts of software systems.
- Parts:
 - **Class diagram**
 - **Use-case diagram**
 - **Activity diagram**
 - **Implementation diagram**

ER vs. UML Class Diagrams

ER Diagram Notation

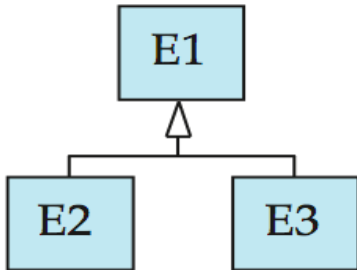
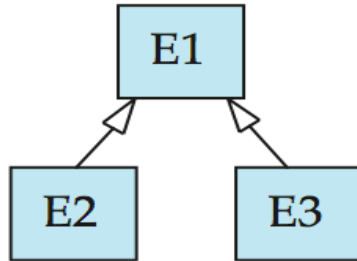
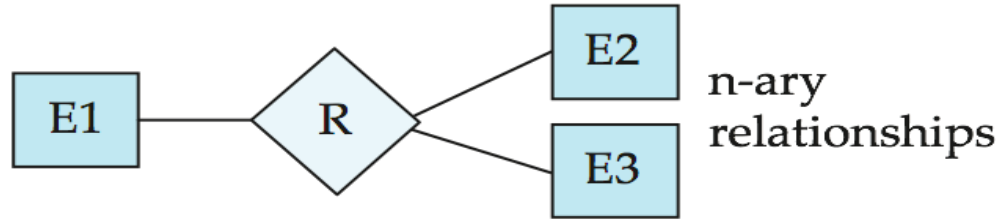


Equivalent in UML



ER vs. UML Class Diagrams

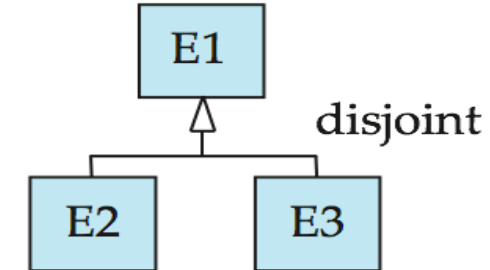
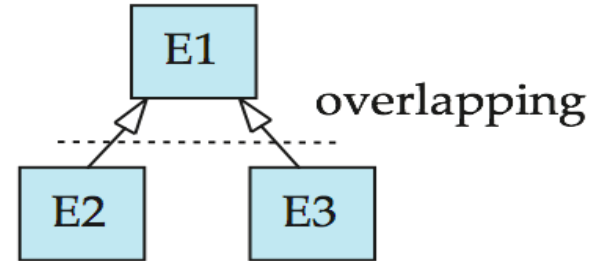
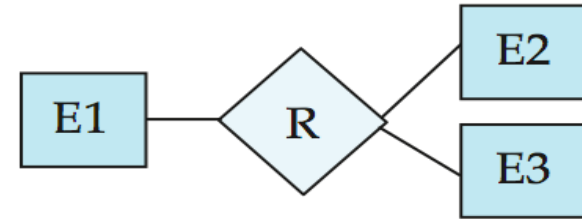
ER Diagram Notation



overlapping
generalization

disjoint
generalization

Equivalent in UML



overlapping

disjoint



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